UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

NATIONAL ENERGY TECHNOLOGY LABORATORY

Learning Spatiotemporal Ultrasonic Guided Wave Propagation in Pipes with Deep Learning Surrogates

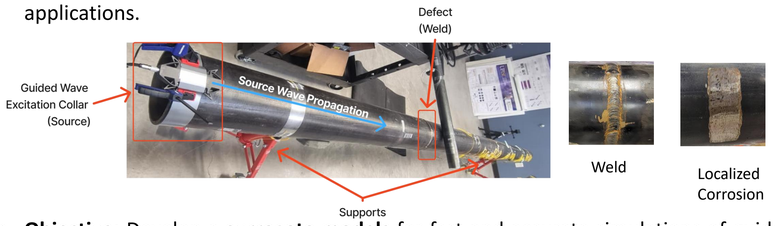
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INTRODUCTION

- **Background:** The degrading **natural gas pipelines** in US prompted the group to develop a **structural health monitoring** system to detect defect in pipelines such as corrosions and welds. **Fiber optics sensors** are used to detect the pipe's response to excited **ultrasonic guided wave** and **train AI models to classify** defects with the aim to deploy it in a **digital twin** platform. [1]
- **Motivation:** One center aspect of this system is a simulation model capable of real-time computation and generalization, which is used (1) to generate extensive data for training the AI classifier and (2) as a digital twin capable making real time inference in visualizing and comparing the data obtained from the sensors. By generalization, we target any pipeline configurations, with kilometers of length, w/o bends, and complex defects resembling real case scenarios as seen in real applications.



- **Objective:** Develop a **surrogate models** for fast and accurate simulations of guided wave response in **arbitrary pipelines** such that axis symmetric guided wave of certain mode is excited, and it propagates along the axis of the pipeline, gets reflected/scattered at various defect present along the way, then finally reflects from the pipe end, and all these waves interfere with each other forming the overall guided wave response which attenuates as it propagates through the pipe.

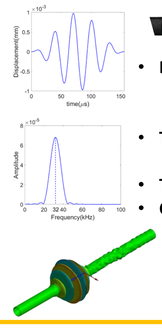
Problem Formulation

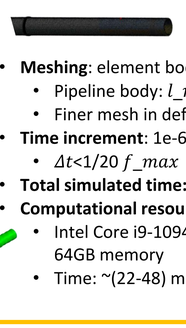
- The surrogate neural model $\mathfrak{S}_\theta$ predicts the updating term on previous state values and next state is updated as [2],
  - $\hat{u}^{n+1} = u^n + \mathfrak{B}(\mathfrak{S}_\theta(u^n, u^{n-1}))$, where $\hat{u}^{n+1}$ is the predicted next step state $\mathfrak{S}_\theta$ is the surrogate model with θ parameters that inputs the current state u^n and previous state u^{n-1}.
  - $\mathfrak{B}$ asserts the boundary condition such that it overwrite the know values at the boundary ∂R of the problem domain R.
- Training is directly done on the predicted change $(\widehat{\Delta u} = \hat{u}^{n+1} - u^n)$ that the surrogate model predicts, such that we chose θ to minimize the loss $E[\left|\left|\widehat{\Delta u} - \Delta u\right|\right|^2]$.
- Then in application the trained model is used recursively in a loop starting from the initial state t_0.

Data Generation

Transient Structural Simulation

- **Solver**: Ansys Mechanical
- **Material**: Structural Steel
- **Domain**: Cylinder (length: 40-65, radius: 1-2.5, and outer thickness: 0.15-0.25), w/o defects
- **Boundary Conditions:**
  - Excitation
    - in a circular axisymmetric surface at one end for first 1.56e-4s
    - Excitation in circumferential direction (Torsional wave T(0,1))
    - 32 kHz 5 cycle sinusoidal signal with a Hanning window
  - Fixes in axial direction at the two ends of pipe

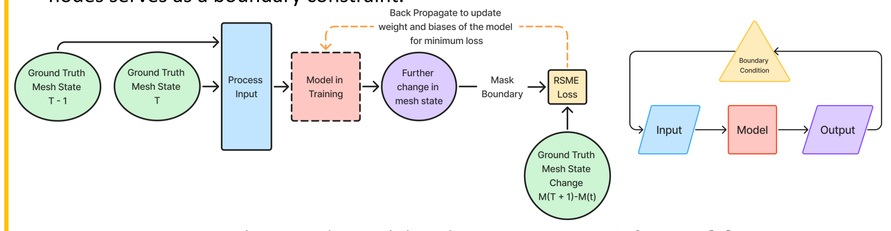




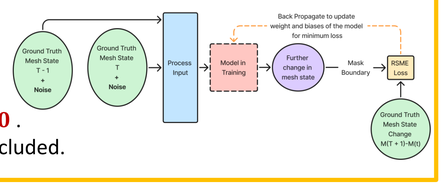
- **Meshing**: element body sizing: 7.62e-3 m
  - Pipeline body: $l_m < \lambda_min/20$
  - Finer mesh in defect region
- **Time increment**: 1e-6s
  - $\Delta t<1/20\ f_max$
- **Total simulated time:** 3.5e-3s
- **Computational resource:**
  - Intel Core i9-10940X @ 3.30GHz 64GB memory
  - Time: ~(22-48) min per simulation

Training and Inference

- **Dynamic Graph Convolution Neural Network(DGCNN)** [3] and **Transolver** [4] (transformer based) network were **trained as updaters**. We **mask the loss** so that this calculation doesn't includes the nodes with defined boundary condition.
- The **inference** process requires a sequential computation at each step. Before predicting the next step, the prediction of the normal nodes is retained but the **positions of the excitation and support (boundary) are overwritten**, where the location of the excitation and support nodes serves as a boundary constraint.



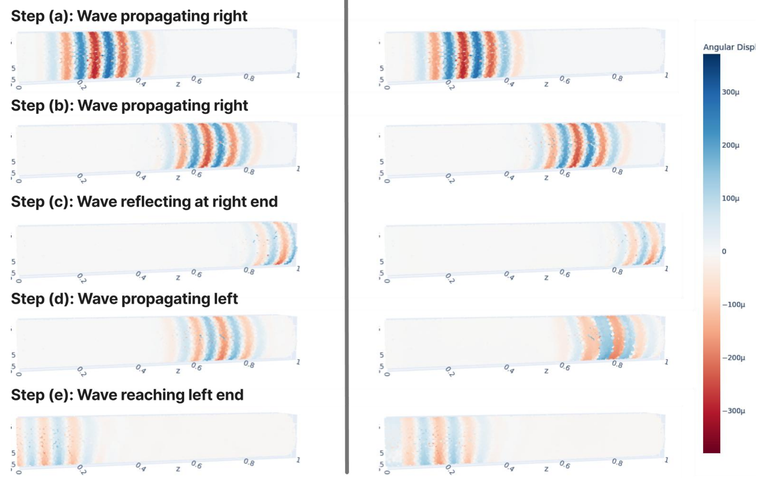
- Noise injection is implemented to stabilize the autoregressive inference.[5]
- Using the previously trained model, the **error in one step prediction** is obtained and the mean and **standard deviation of error** at all the points in all steps of all trajectories.
- Then the obtained standard deviation is used to generate a **random gaussian noise** $\mathfrak{N}(\mathbf{0}, \boldsymbol{\sigma})$ of zero mean, which is added to the training data, both input positions and output positions as augmented as
  - Input positions:
    - $\boldsymbol{u_{t-i} = u_{t-i} + \mathfrak{N} \mid \forall\, i = 0, 1, 2, ..\mid}$
  - and Output positions:
    - $\boldsymbol{u_{t+1} = u_{t+1} + (1-\gamma)\mathfrak{N} \mid \gamma = 0.1 \sim 0}$.
  - This steps help in preserving the physics included.



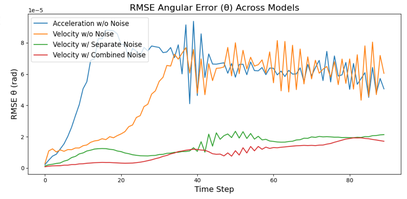
PRELIMINARY RESULTS & PERSPECTIVES

Ground Truth: ANSYS Mechanical

Prediction: Transolver



- All three-directional deformation: **radial, circumferential, and axial** were predicted, while the most dominant circumferential direction results are shown.
- Surrogate model effectively captured the reflection from far ends, reflection from defects, and attenuation of waves as it propagated forward.



References

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[4] Wu, Haixu, et al arXiv preprint arXiv:2402.02366 (2024).

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ACKNOWLEDGEMENT

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